



CampusBite — Student Food Ordering App

A Flutter mobile application that lets university students browse the campus cafeteria menu, place food orders in advance, and skip the queue. Built with **Flutter** and **Firestore**.

This repository (foodorder) contains the **Student App**. There is the admin companion app which will not be published.

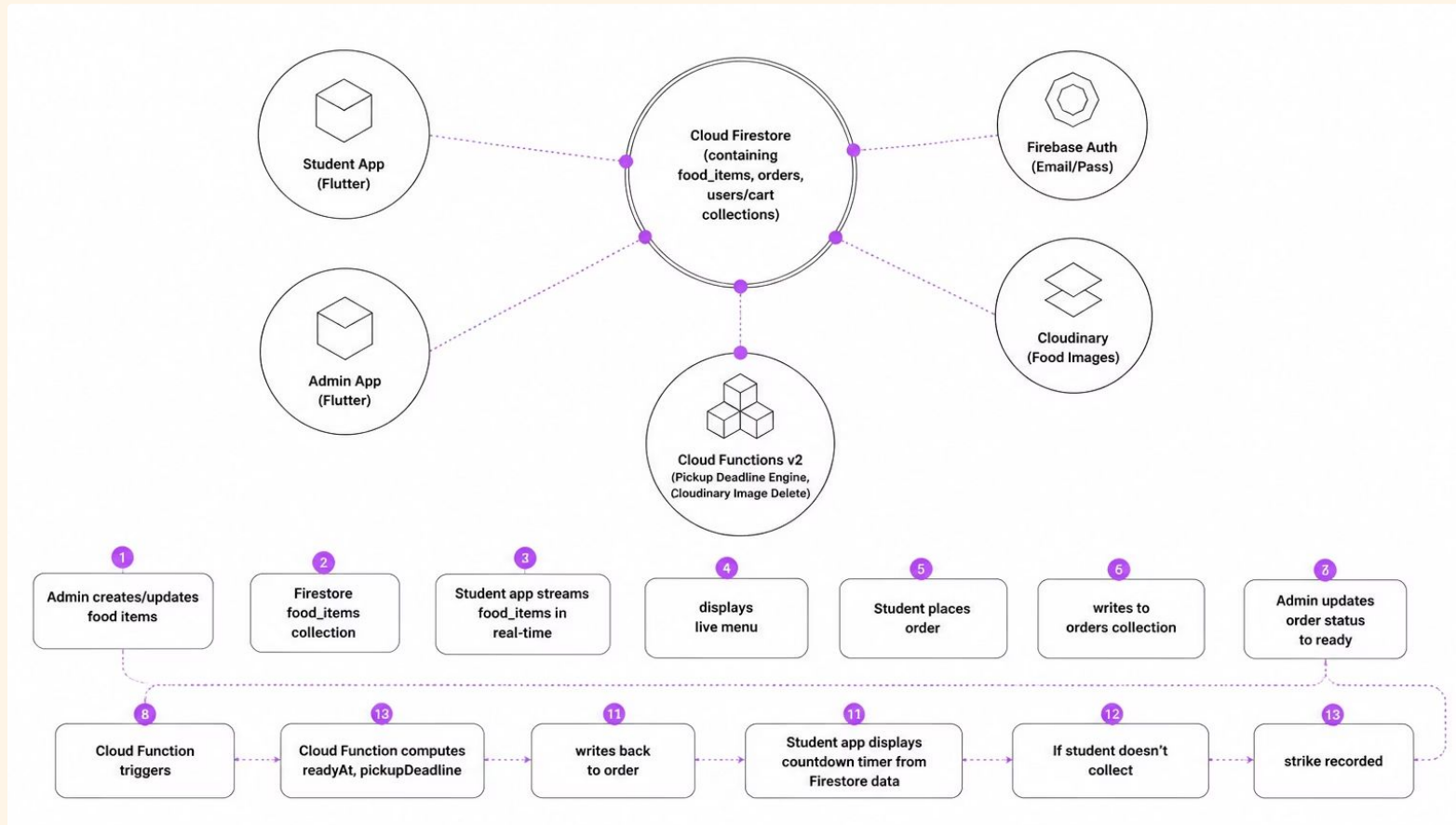
Project Overview

CampusBite solves common problems at university cafeterias:

Problem	Solution
Long queues at meal times	Students order in advance from their phone
No visibility into today's menu	Live menu synced from Firestore in real-time
Abandoned orders causing food waste	Pickup deadline engine with countdown timers
No accountability for no-shows	Automated strike system for missed pickups

The system consists of **three components** only the foodorder student app is in this directory but all two share a single Firebase project:

System Architecture



Data Flow Steps

1. Admin creates/updates food items → Firestore `food_items` collection
2. Student app streams `food_items` in real-time → displays live menu
3. Student places order → writes to `orders` collection
4. Admin updates order status to "ready" → Cloud Function triggers
5. Cloud Function computes `readyAt`, `pickupDeadline` → writes back to order
6. Student app displays countdown timer from Firestore data
7. If student doesn't collect → strike recorded

App Navigation

The student app uses `go_router` with authentication-aware redirects:

<code>/</code>	WelcomeScreen (unauthenticated landing)
<code>/register</code>	RegisterScreen
<code>/login</code>	LoginScreen
<code>/main</code>	MainScreen (authenticated, contains bottom nav)
<code>/terms</code>	TermsScreen
<code>/support</code>	SupportScreen
<code>/support/faq</code>	FaqScreen
<code>/support/contact</code>	ContactScreen

Bottom Navigation Bar

Index	Tab	Screen
0	Home	HomeScreen — featured items, sections, food feed
1	Categories	CategoryScreen — browse by Breakfast, Lunch, etc.
2	Search	SearchBarScreen — Firestore prefix-based search
3	Orders	OrdersScreen — active orders with countdown, history
4	Account	AccountScreen — profile, strikes, settings, logout

Key Features

For Students

- **Registration & Login** — Email/password authentication via Firebase Auth.
- **Browse Menu** — Real-time Firestore stream of available food items sorted by category.
- **Categories** — Quick navigation to browse food items by categories (e.g. Breakfast, Lunch, Dinner, Teasers, Drinks).
- **Search** — Fast, prefix-based Firestore search with 300ms debounce and in-memory caching.
- **Cart & Checkout** — Add items, choose target cafe, and review total amount before placing orders.

For Cafeteria Admins

- **New Order Alerts** — Automatic notification of incoming student orders in real time.
- **Order Flow Manager** — Control order stages from pending, accepted, preparing, to ready for collection.
- **Automatic Strike Engine** — Automated detection of missed pickup deadlines that flags no-shows, issues strikes, and suspends repeat offenders.
- **Admin Pardon/Reactivation** — Transactional tools to pardon strikes, reset counts, and lift user suspensions, fully logged via audit trails.

Key Features

For Students

- **Orders & Countdown Timers** — Track active order statuses and see real-time pickup countdown timers synced with server-enforced deadlines.
- **Student Discipline Card** — Transparent in-app view of current strikes and suspension status.
- **Notification Center** — In-app notification feed supporting real-time alerts for order status changes, pickup reminders, strike actions, and account suspension events.
- **Distance-Based Pickup Window** — Automatically calculates walking distance to target cafe using GPS/Geolocator and requests corresponding pickup time window at checkout (from 10 to 25 minutes) to ensure freshness.

Search System

The search feature uses a **prefix-based array-contains** strategy for Firestore:

1

When admin
creates/updates a food item

SearchHelper auto-generates:

- `titleLower` — lowercase title
- `keywords` — unique lowercase words from title, category, and dietary tags

2

When a student searches
SearchService:

- Trims and lowercases the query
- Checks an in-memory cache before hitting Firestore
- Queries `food_items` where `available == true` AND `searchPrefixes` array-contains query

3

The search UI
(SearchBarScreen)

Debounces input by 300ms and displays results with food images, price, category, and cafe info.

Distance-Based Pickup Deadline & Location-Based Calculations

To minimize food waste from abandoned orders while keeping pickup schedules fair for students, CampusBite calculates a dynamic pickup window based on the student's walking distance to the cafeteria at the moment they checkout:

1 GPS Distance Calculation

When the student clicks "Place Order", the app requests temporary, single-use access to device location. It computes the straight-line walking distance (in meters) between the student's coordinates and the selected cafe's geographic coordinates using the `Geocator.distanceBetween` API.

2 Dynamic Pickup Windows

The calculated distance maps to specific pickup windows via the `PickupWindowService`:

- **0 – 250 meters:** 10-minute pickup window
- **251 – 600 meters:** 15-minute pickup window
- **601 – 1200 meters:** 20-minute pickup window
- **Above 1200 meters:** 25-minute pickup window

Location-Based Calculation

1 Privacy Protections

To protect student privacy, the app **never** uploads, stores, or transmits precise GPS coordinates or location history to Firestore or any external server. Location data is calculated strictly client-side. Only the resulting distance (in meters) and the pickup window (in minutes) are persisted inside the order document under `distanceMeters` and `pickupWindowMinutes`.

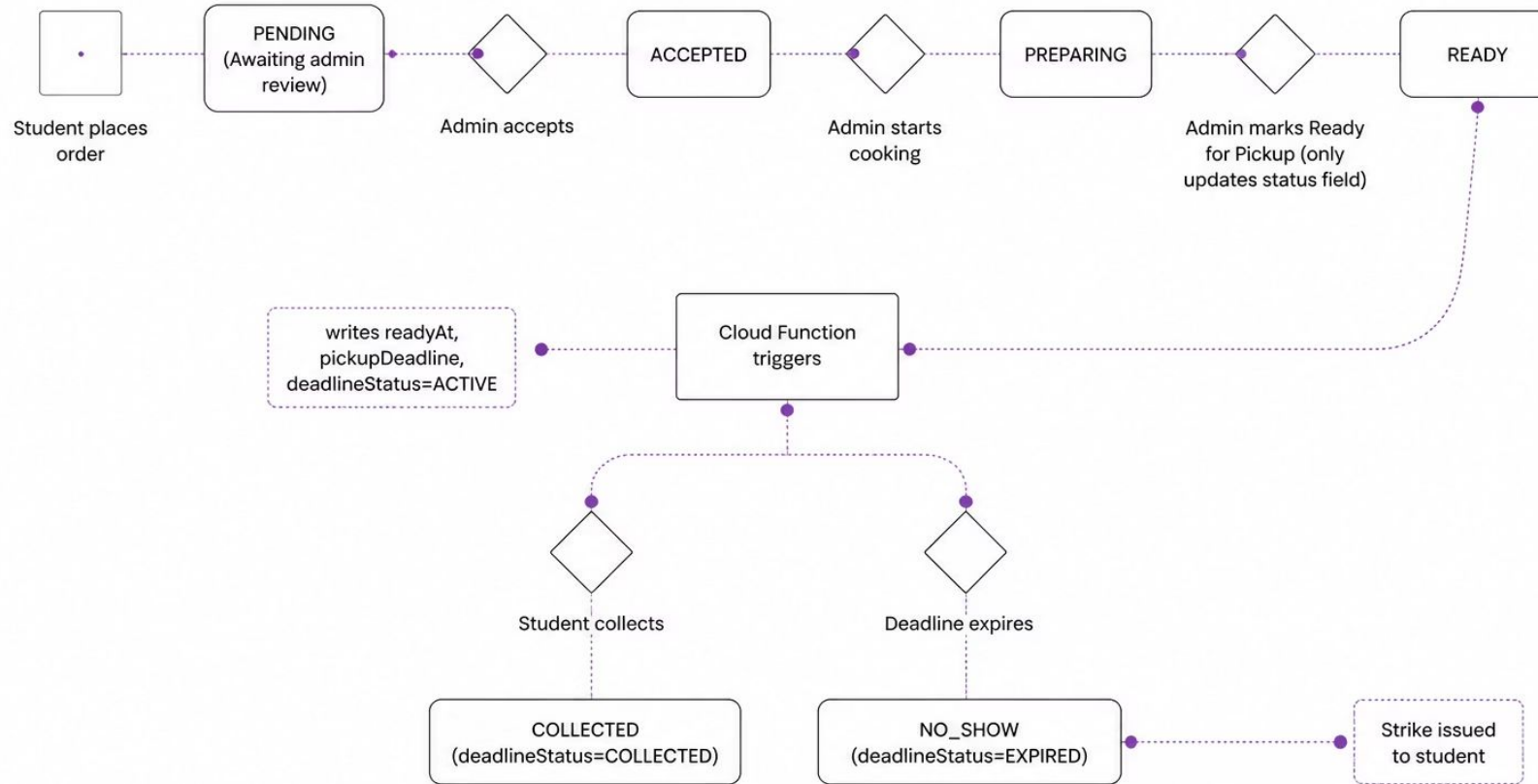
2 Deadline Enforcement

When a cafeteria admin marks the order's status as "ready", the `onOrderStatusChanged` Cloud Function reads the stored `pickupWindowMinutes` from the order document and atomically calculates the deadline:

$$\text{pickupDeadline} = \text{readyAt} + (\text{pickupWindowMinutes} \times 60 \text{ seconds})$$

The function updates `pickupDeadline` and sets `deadlineStatus` to "ACTIVE". If `pickupWindowMinutes` is missing, it defaults to a 20-minute fallback.

Order Lifecycle



Student Discipline & Automatic Strike Engine

CampusBite enforces pickup accountability with an automated discipline system that marks missed orders, limits repeat offenses, and lets administrators restore accounts when needed.

1. Automatic Strike Engine

- A scheduled Cloud Function checks for orders that were ready but not collected before the pickup deadline.
- Expired orders are marked `no_show` and the student receives a strike.
- Repeated no-shows can raise the strike count to the suspension threshold.
- Students are notified when a strike is issued, and again if their account becomes suspended.

2. Suspension Enforcement

When a student reaches the suspension threshold, the account is blocked from placing new orders or adding items to the cart.

- `isAccountSuspended()` prevents further ordering.
- The Account Screen shows a suspension banner.

3. Administrative Actions & Pardons

Administrators can review and adjust strike status in the Admin App.

Production Notification Platform

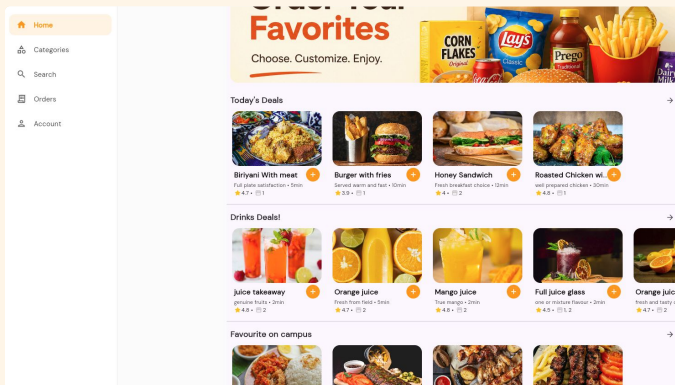
CampusBite uses a scalable notification layer that keeps delivery decoupled from core business logic.

1. Architectural Flow

2. Key Features

- **Duplicate Prevention:** Each business event gets a unique `eventId`. `NotificationService` checks for an existing notification first, so each event is written exactly once.
- **Decoupled Logic:** Notifications react to business events only; they never block or drive core app behavior.
- **Real-Time Streams:** Student and admin apps listen to their own notification stream for instant updates.
- **Soft Delete:** Deleted notifications are flagged out of the UI instead of being removed immediately.
- **Deep Linking:** Tapping a notification takes users straight to the relevant screen, such as an order, account, or strike history.

Website.



foodapp.larason.space

Campus Bite

Order your university meal at your fingertips

